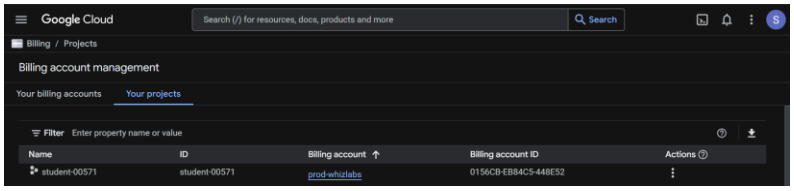
Capstone Project: Enterprise Google Cloud Infrastructure Deployment

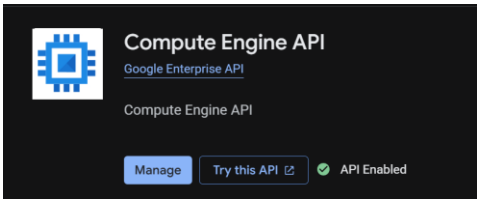
1. **Project Setup**
   - Create “NEW PROJECT”
   - Select the Project
   - Verify Billing



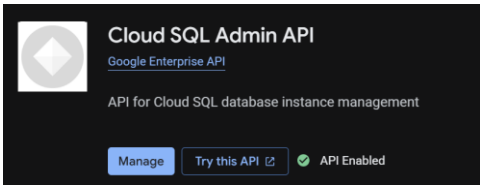
   - Set the default region and zone using Activate Cloud Shell

   - Enable Required APIs

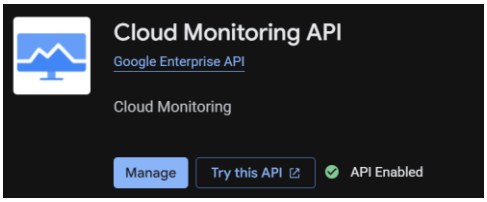
     Enable these:
     - Compute Engine API



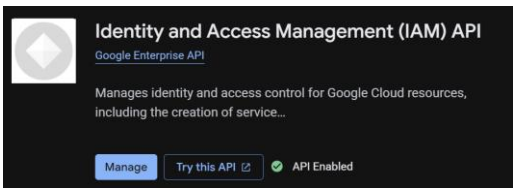
     - Cloud SQL Admin API



     - Cloud Monitoring API



     - Identity and Access Management API

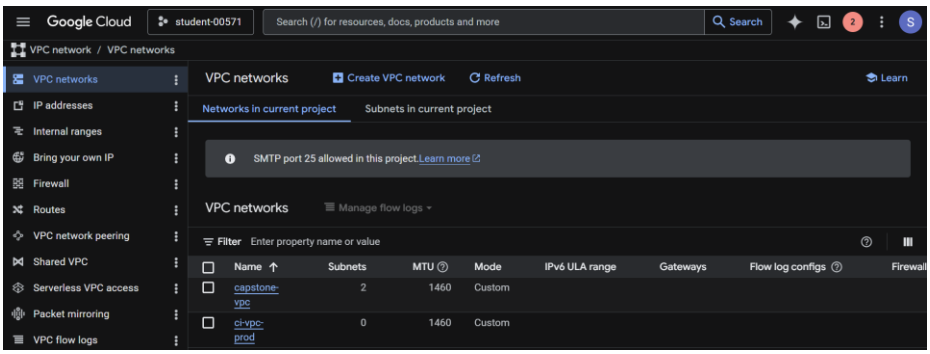


2. Networking Layer (VPC Design)

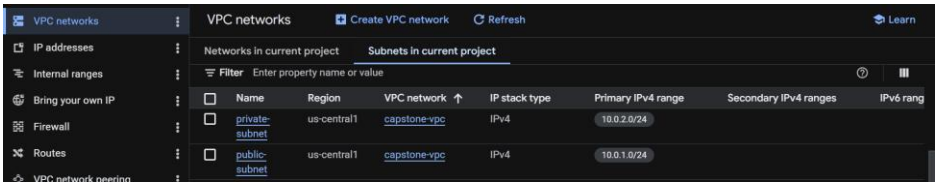
A custom VPC was created using Virtual Private Cloud.

- Two subnets created:
  - Public subnet (10.0.1.0/24)
  - Private subnet (10.0.2.0/24)
- Firewall rules configured to allow HTTP traffic and internal communication
- Cloud NAT enabled to allow private instances to access the internet securely

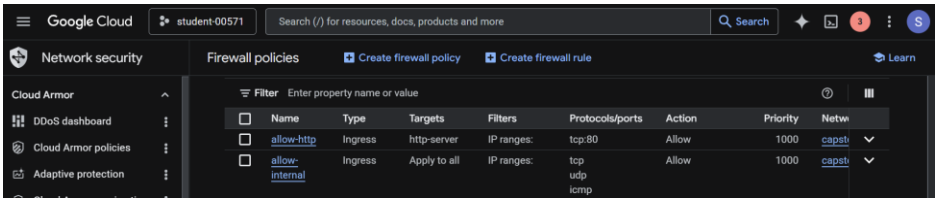
- Create VPC



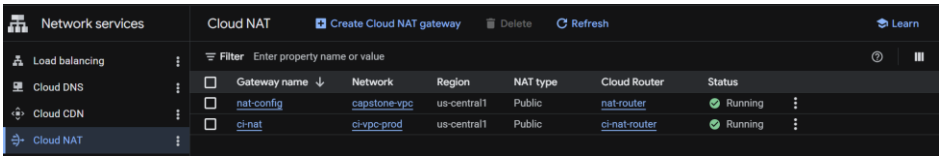
- Create Subnets



- Create Firewall Rule



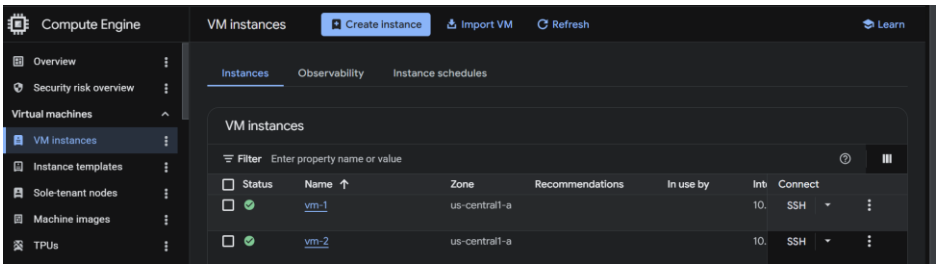
- Cloud NAT



3. Compute Layer

Two VMs (**vm-1** and **vm-2**) were created using Compute Engine in a private subnet.

- No external IP (not publicly exposed)
- Nginx installed on both VMs
- Used for backend web servers



4. Load Balancing

An HTTP Load Balancer was created using Cloud Load Balancing to distribute traffic across two backend VMs (vm-1 and vm-2).

- Instance group created with both VMs
- Health check configured on port 80
- Backend service linked to the instance group
- Public IP provided for external access

```
student373767701@cloudshell:~ (student-00571)$ gcloud compute instance-groups unmanaged create web-group \
  --zone=us-central1-a
Created [https://www.googleapis.com/compute/v1/projects/student-00571/zones/us-central1-a/instanceGroups/web-group].
NAME: web-group
LOCATION: us-central1-a
SCOPE: zone
NETWORK:
MANAGED:
INSTANCES: 0
student373767701@cloudshell:~ (student-00571)$ gcloud compute instance-groups unmanaged add-instances web-group \
  --instances=vm-1,vm-2 \
  --zone=us-central1-a
Updated [https://www.googleapis.com/compute/v1/projects/student-00571/zones/us-central1-a/instanceGroups/web-group].
student373767701@cloudshell:~ (student-00571)$
```

```
student373767701@cloudshell:~ (student-00571)$ gcloud compute health-checks create http http-health-check \
  --port 80
Created [https://www.googleapis.com/compute/v1/projects/student-00571/global/healthChecks/http-health-check].
NAME: http-health-check
PROTOCOL: HTTP
student373767701@cloudshell:~ (student-00571)$ gcloud compute backend-services create web-backend \
  --protocol=HTTP \
  --health-checks=http-health-check \
  --global
Created [https://www.googleapis.com/compute/v1/projects/student-00571/global/backendServices/web-backend].
NAME: web-backend
BACKENDS:
PROTOCOL: HTTP
student373767701@cloudshell:~ (student-00571)$ gcloud compute backend-services add-backend web-backend \
  --instance-group=web-group \
  --instance-group-zone=us-central1-a \
  --global
Updated [https://www.googleapis.com/compute/v1/projects/student-00571/global/backendServices/web-backend].
student373767701@cloudshell:~ (student-00571)$ gcloud compute url-maps create web-map \
  --default-service web-backend
Created [https://www.googleapis.com/compute/v1/projects/student-00571/global/urlMaps/web-map].
NAME: web-map
DEFAULT SERVICE: backendServices/web-backend
```

```
student373767701@cloudshell:~ (student-00571)$ gcloud compute forwarding-rules create http-rule   --global   --target-http-proxy=http-proxy   --ports=80
ERROR: (gcloud.compute.forwarding-rules.create) Could not fetch resource:
 - The resource 'projects/student-00571/global/targetHttpProxies/http-proxy' was not found

student373767701@cloudshell:~ (student-00571)$ gcloud compute target-http-proxies create http-proxy \
  --url-map=web-map
Created [https://www.googleapis.com/compute/v1/projects/student-00571/global/targetHttpProxies/http-proxy].
NAME: http-proxy
URL MAP: web-map
student373767701@cloudshell:~ (student-00571)$ gcloud compute forwarding-rules create http-rule \
  --global \
  --target-http-proxy=http-proxy \
  --ports=80
Created [https://www.googleapis.com/compute/v1/projects/student-00571/global/forwardingRules/http-rule].
student373767701@cloudshell:~ (student-00571)$
```

5. Data Layer

A managed database layer using Cloud SQL (MySQL) was designed as part of the architecture. However, due to sandbox IAM restrictions, database provisioning was not available in this environment. The design includes a private, secure database intended to be accessible only from internal VPC resources, following least privilege and security best practices.

6. **Security**
   Firewall rules were configured to allow HTTP traffic on port 80. Virtual machines were deployed without external IP addresses to ensure secure private access only through the load balancer.
7. **Monitoring & Logging**
   Cloud Monitoring and Cloud Logging were enabled to track system performance and logs. An alert policy was created to notify when CPU utilization exceeds 80%, ensuring proactive monitoring of the infrastructure.
8. **Automation (Mandatory)**
   Terraform was used to automate infrastructure deployment. A VPC, public and private subnets, and firewall rules were provisioned as Infrastructure as Code, ensuring repeatable and scalable deployment on Google Cloud.